ABDULLAH KAISAR FARDIN

AI Developer

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI developer specializing in generative AI systems, multimodal pipelines, and scalable backend engineering — with hands-on production experience building Hybrid RAG systems, agentic workflows, and real-time AI applications.

Skilled in designing high-concurrency architectures using Django 5, FastAPI, Celery, and WebSockets, with a strong focus on performance, security, and production deployment.

Proven track record of shipping AI-powered products that integrate LLMs, computer vision, and 3D data pipelines — delivering reliable, scalable solutions across 5+ production platforms.

WORK EXPERIENCE

Jr AI Developer — Aug 2025 – Present

SparkTech Agency, Dhaka

- Architected modular monolith backends using Django 5 and FastAPI, implementing custom JSON renderers and exception handlers to enforce consistent API contracts across 5+ production AI platforms.
- Engineered high-concurrency AI pipelines by splitting Celery worker pools into Gevent (I/O-bound AI tasks) and Prefork (CPU-bound tasks), ensuring 99.9% system availability during peak multimodal processing.
- Developed a Hybrid KG-RAG system integrating Neo4j (Graph DB) and Pinecone (Vector DB), utilizing Natural Language Inference (NLI) to atomize and verify LLM claims, significantly reducing hallucination rates.
- Implemented secure, stateless authentication using a JWT + OTP hybrid flow, incorporating password breach-check APIs (PwnedPasswords) and request throttling to harden platform security.
- Optimized 3D data pipelines using Trimesh and NumPy, automating anisotropic scaling and geometric landmark estimation for biometric analysis from 2D image uploads.

Trainee AI Developer — May 2025 – Aug 2025

SparkTech Agency, Dhaka

- Developed proof-of-concept AI scripts and Jupyter notebooks for Computer Vision (MediaPipe) and NLP, focusing on facial landmark mesh analysis and text preprocessing pipelines.
- Streamlined development workflows by containerizing Python environments with Docker, ensuring parity between local research notebooks and cloud-deployed services.
- Conducted research on LLM context management, implementing sliding-window strategies to optimize token usage and response relevance for early-stage chat prototypes.

TECHNICAL SKILLS

Generative AI & LLMs: OpenAI API (GPT-4o/Vision), Hybrid RAG (Vector + Knowledge Graph), Whisper (STT), ElevenLabs (TTS), Agentic Workflows, Prompt Engineering, NLI-based Hallucination Mitigation

Computer Vision & 3D: MediaPipe (Facial Mesh), OpenCV, Trimesh (3D Geometric Analytics), 3D Mesh Morphing, Anisotropic Scaling, Image Inpainting, Visual Grounding

Core Frameworks: Django 5, Django REST Framework (DRF), FastAPI (Async), REST API, Celery, Django Channels (WebSockets)

Data Engineering: Neo4j (Graph DB), Pinecone (Vector DB), Redis (Caching/Broker), PostgreSQL (Indexing/Optimization), MongoDB (Motor), S3

Languages & Science: Python (Advanced/Production), SQL, Cypher (Neo4j), NumPy, Pandas, Scikit-Learn, Graph Theory (Centrality Scoring)

DevOps & Infrastructure: Docker & Docker Compose, Nginx (Reverse Proxy), ASGI/WSGI (Uvicorn/Gunicorn), Git, CI/CD, FCM (Push Notifications), RevenueCat

PROJECTS

Explainable Hybrid KG-RAG Assistant

Tech: *FastAPI, Neo4j, Pinecone, LLMs, NLI* | *Private repository — client project*

- Built a Hybrid RAG (Retrieval-Augmented Generation) system combining vector search (Pinecone) and knowledge graph traversal (Neo4j) to improve multi-hop reasoning and retrieval accuracy.
- Implemented a post-response validation layer using Natural Language Inference (NLI) to break LLM outputs into atomic claims and verify them against retrieved sources, reducing hallucinations.
- Applied graph algorithms (Degree Centrality) to rank entity importance and improve answer grounding for complex queries.

Wingman (AI Coaching System)

Tech: *Django, Channels, Redis, Celery, OpenAI (Vision + Whisper)* | *Private repository — client project*

- Built a multimodal AI assistant supporting text, image, and audio inputs using OpenAI Vision and Whisper APIs.
- Designed real-time chat system with WebSockets (Django Channels) and background workers for OCR, transcription, and AI processing.
- Improved scalability with distributed worker queues and caching, ensuring responsive UX under high concurrency.
- Implemented secure authentication (JWT, OAuth, OTP) and production deployment with Docker, Nginx, PgBouncer.

MagicTale AI Backend

Tech: *GPT-4o, DALL·E, ElevenLabs, Django, Celery* | *Private repository — client project*

- Built an end-to-end AI storytelling pipeline combining text generation, image creation, and voice synthesis.
- Implemented asynchronous task orchestration (Celery chains) to manage long-running media generation workflows.
- Designed real-time progress updates via WebSockets, improving user experience during content generation.
- Added subscription management and API security controls for scalable SaaS deployment.

EDUCATION

MSc in Data Science & Analytics

East West University

Jan 2026 – Present (Expected Apr 2027)

BSc in Computer Science & Engineering

Bangladesh University of Business & Technology

2019 – 2024